

Geothermal Energy Usage Comparison

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Purpose

This summary uses the Energy Saver North Carolina audit as the baseline and compares the expected annual energy usage for two quoted geothermal packages: a 2-ton WaterFurnace 7 Series package and a 2.5-ton WaterFurnace 5 Series package. The estimate also includes a desuperheater piped into the existing 50-gallon electric water heater and compares open-loop versus horizontal closed-loop operation.

Audit baseline

Audit item	Value
Modeled current electric usage	10,101 kWh/year
Modeled current wood usage	1.09 cords/year
Modeled current total energy cost	\$1,854/year
Audit electric rate	\$0.1511/kWh
Audit wood cost	\$300/cord
Existing heat pump size	30,000 Btu/h, or 2.5 tons
Heating design load	24,661 Btu/h
Cooling load	24,132 sensible Btu/h + 1,629 latent Btu/h; approximately 25,761 total Btu/h

Source: Energy Saver North Carolina Home Energy Assessment, audit dated March 26, 2026. Figures are weather-normalized modeled values, not necessarily exact historic utility usage.

Estimated annual energy cost by option

The desuperheater is treated as hot-water assist only. It is not assumed to fully replace the existing water heater. Estimated water-heating savings are included in the geothermal options below.

Scenario	Loop type	Est. annual electric use	Wood use	Est. annual energy cost	Savings vs. audit baseline
Current audit baseline	Existing air-source / wood / electric tank	10,101 kWh	1.09 cords	~\$1,854	-
2-ton WaterFurnace 7 Series + desuperheater	Horizontal closed loop	~6,450-6,850 kWh	0 cords	~\$975-\$1,035	~\$820-\$880/year
2-ton WaterFurnace 7 Series + desuperheater	Open loop	~6,250-6,700 kWh	0 cords	~\$945-\$1,010	~\$845-\$910/year
2.5-ton WaterFurnace 5 Series + desuperheater	Horizontal closed loop	~6,650-7,050 kWh	0 cords	~\$1,005-\$1,065	~\$790-\$850/year
2.5-ton WaterFurnace 5 Series + desuperheater	Open loop	~6,450-6,900 kWh	0 cords	~\$975-\$1,045	~\$810-\$880/year

Main takeaway

With a desuperheater tied into the existing 50-gallon electric water heater, the expected annual energy cost drops from the audit's modeled \$1,854/year to roughly \$945-\$1,065/year, depending on model selection and loop type. This equals an estimated savings range of about \$790-\$910/year, or roughly \$65-\$75/month averaged over the year.

Desuperheater impact

Item	Estimated impact
Existing water heater	50-gallon electric tank water heater retained
Desuperheater role	Hot-water assist only; not a full replacement for the water heater
Estimated annual water-heating offset	~500-800 kWh/year
Estimated dollar savings	~\$75-\$120/year at \$0.1511/kWh

The desuperheater usually helps most during cooling season, when heat is being rejected from the home. In heating season it can still help, but the heat is ultimately being produced by the geothermal system, so the benefit is not the same as free summer heat recovery.

Open loop vs. horizontal closed loop

Comparison point	Expected difference
Open-loop operating advantage	Typically modest because entering water temperatures can be very stable
Estimated HVAC kWh reduction vs. horizontal loop	~3%-8% before considering pumping details
Estimated net annual electric difference	~150-400 kWh/year
Estimated annual dollar difference	~\$25-\$60/year at the audit electric rate
Main deciding factors	Water quality, well capacity, discharge method, local permitting, maintenance risk, and long-term reliability

Open loop should not be oversold as a large energy saver for this project. The energy difference is likely smaller than the difference in installation requirements, water management, and maintenance risk. If the well water quality and discharge setup are excellent, open loop can be attractive. If not, a horizontal closed loop is often the cleaner long-term ownership story.

Sizing and package notes

The audit load is close to 2 tons, but the home's existing air-source system is 2.5 tons. The 2-ton 7 Series package may be appropriate because of its variable-capacity operation, but its exact output should be confirmed against the heating and cooling design loads at the expected entering water temperature. The 2.5-ton 5 Series package is the safer capacity match to the existing system size and audit cooling load, though it may not provide the same modulation and part-load comfort as the 7 Series.

Homeowner-facing wording

Based on the energy audit, switching from the existing air-source heat pump, wood heat, and electric water heater setup to a WaterFurnace geothermal system with desuperheater should reduce annual energy cost from about \$1,854 per year to roughly \$945-\$1,065 per year. That is about \$790-\$910 per year in estimated savings. The open-loop option may save another \$25-\$60 per year compared with a horizontal closed loop, but the final decision should be based more on water quality, well capacity, discharge requirements, permitting, and long-term reliability than on the small energy difference alone.

Recommended positioning

Recommend the 2-ton 7 Series if the load calculation, unit capacity tables, loop design, and auxiliary heat strategy confirm that it will cover the home properly at design conditions. Recommend the 2.5-ton 5 Series if the customer wants the safer sizing match and likely lower installed cost. Both options should materially reduce annual energy use compared with the audit baseline, especially if wood use is eliminated and the desuperheater offsets a portion of electric water heating.

Notes: This is a budget-level comparison for discussion only. Final usage depends on equipment submittals, actual loop design, entering water temperatures, pumping watts, thermostat settings, auxiliary heat use, duct performance, domestic hot-water usage, and homeowner operating behavior.